A=5
M=9
M
↓
0x30AA

8. STORE T1
LOAD 0x30AA
STORE T2
LOAD T1
STORE 0x30AA
LOAD T2

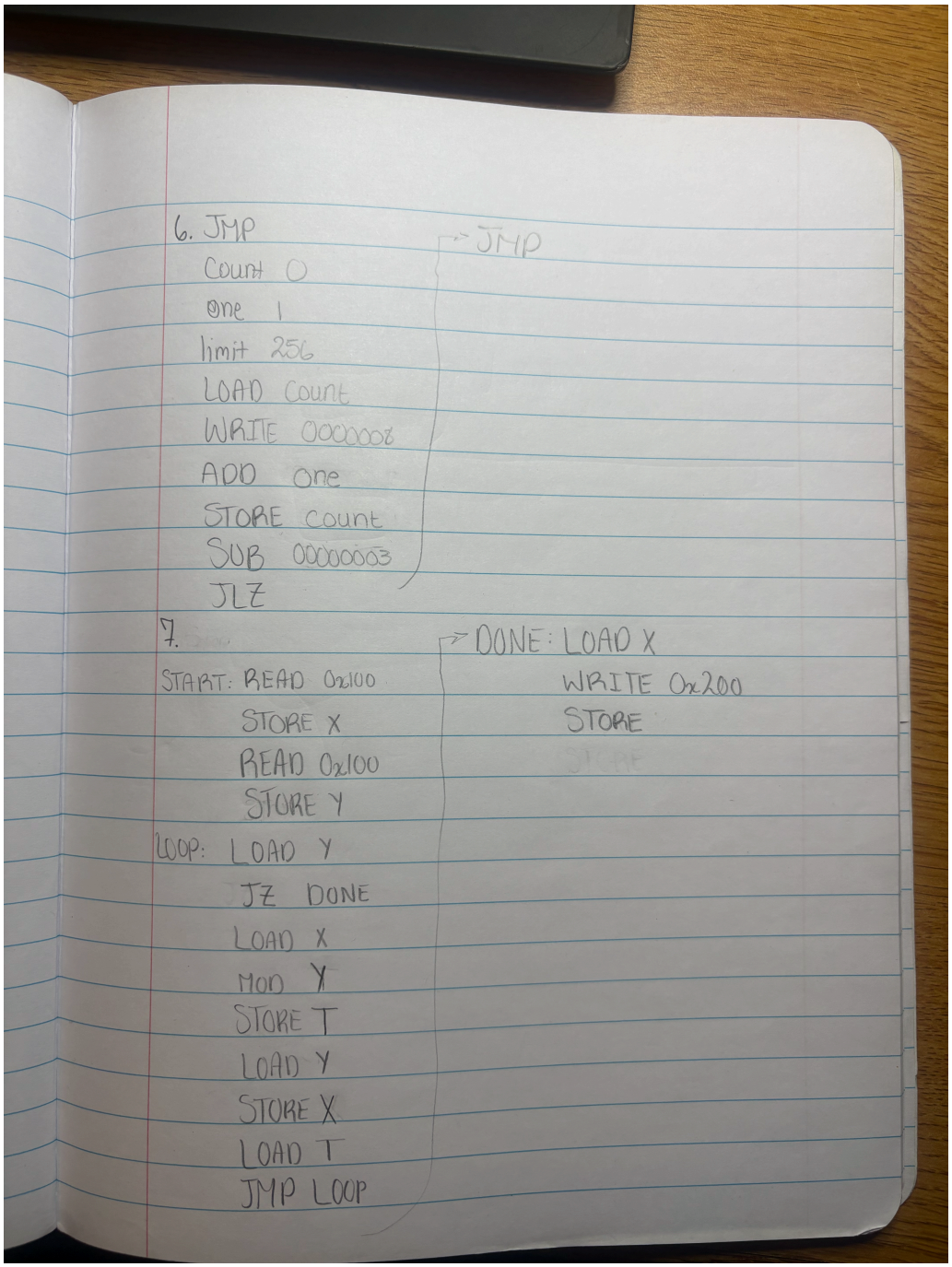
STORE T
XOR 0x30AA
STORE 0x30AA
XOR T

→ Continue
STORE T
LOAD 0x30AA
XOR T
STORE 0x30AA
LOAD T

9.) JZ 0x837BBE1
JGE 0x837BBE1

PART 1 10.) The program notices that the sequence swaps the contents in r8 and r9. And after the instructions r8 now holds r8's original value

PART 2 This effect happens because of XOR which inverts itself. Because XOR took the same value, the two would cancel out, as XOR compares two bits and gives 1 if they are different and 0 if they are the same.



6. JMP

```
Count 0
One 1
limit 256
LOAD Count
WRITE 0000008
ADD One
STORE Count
SUB 00000003
JLZ
```

JMP

7.

```
START: READ 0x100
       STORE X
       READ 0x100
       STORE Y
LOOP:  LOAD Y
       JZ DONE
       LOAD X
       MOD Y
       STORE T
       LOAD Y
       STORE X
       LOAD T
       JMP LOOP
```

```
DONE: LOAD X
      WRITE 0x200
      STORE
```